Lab 14 – 30-12-2024

Note: The code to create adjacency list You can use the code to create adjacency list.

Task 1: Write a program that identifies and prints all pairs of nodes in an undirected graph that are separated by exactly k edges. You are given an undirected graph where nodes are connected by edges. Your task is to find and print all pairs of nodes that have an exact shortest path of k edges between them.

Input Format:

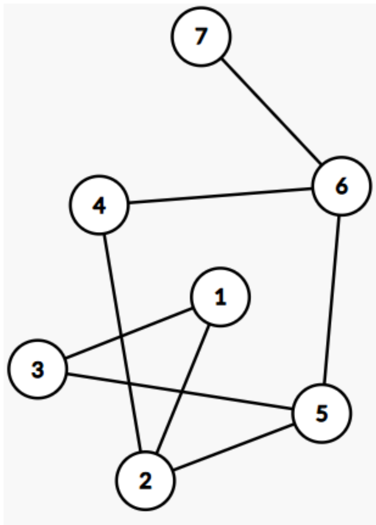
Number of nodes (N) and edges (E). N pairs of integers representing the edges in the graph. A positive integer k, the required edge distance.

Output Format:

Print all pairs of nodes (u, v) such that the shortest distance between u and v is exactly k edges.

Sample Input:

6 7 A vector containing all edges
1 2
1 3
2 4
2 5
3 5
4 6
5 6
2 k



Sample Output:
1 4
1 5
2 3
2 6
3 6
4 5
4 7
5 7

Explanation:

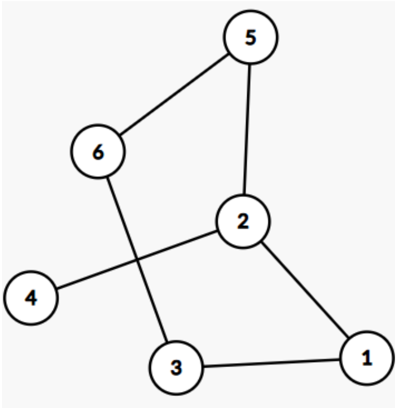
In the given graph:

Node 1 is exactly 2 edges away from nodes 4 and 5.
Node 2 is exactly 2 edges away from node 6.
Node 3 is exactly 2 edges away from node 6.

Task 2: Write a program to find all leaf nodes in a given tree. A leaf node is a node with a degree of 1.

Input Format:

- Number of nodes N and edges E.
- E pairs of integers representing the edges in the tree.



Output Format: Print all leaf nodes in the tree.

Sample Input:

6 5
1 2
1 3
2 4
2 5
3 6

Sample Output:

4 5 6

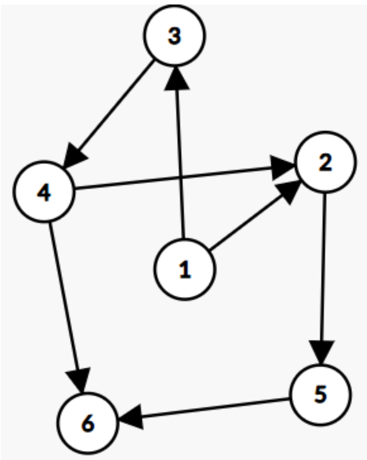
Task 3: Write a program to find the length of the longest path in a directed acyclic graph (DAG).

Input Format:

- Number of nodes N and edges E.
- E pairs of integers representing the edges in the DAG.

Output Format: Print the length of the longest path in the DAG.

Sample Input:



6 6
1 2
1 3
3 4
2 4
2 5
4 6

Sample Output:

4

Explanation:

In the given graph, the longest path is from node 3 to node 6 via 3->4->2->5->6

Node 1 is exactly 2 edges away from nodes 4 and 5.
Node 2 is exactly 2 edges away from node 6.
Node 3 is exactly 2 edges away from node 6.